

- Business Requirements Document (BRD)**
- Project Title:** Data Integration and Aggregation for Customer and Order Data
- Introduction:**

The purpose of this project is to design and implement a data integration and aggregation process for customer and order data stored in CSV files. The process involves loading the data into Delta tables, performing data cleansing, creating an order summary table, and aggregating customer spend data.

- Business Requirements:**

1. **Data Ingestion**

- Read source CSV data from the specified volumes:
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`.
- Load the data into Delta tables `customer` and `order`.

2. **Data Cleansing**

- Remove "Null"/Null and duplicate records from both `customer` and `order` tables.

3. **Order Summary Table Creation**

- Create an `ordersummary` table if it does not exist in `catalog="gen\_ai\_poc\_databrickscoe"` and `schema="sdlc\_wizard"`.
- The schema of the `ordersummary` table should be: `CustId`, `Name`, `EmailId`, `Region`, `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`.

4. **Data Integration**

- Join `customer` and `order` data using the `CustId` field.
- Load the joined data into an SCD Type 2 table under `catalog=gen\_ai\_poc\_databrickscoe`, `schema=sdlc\_wizard`, `table=ordersummary`.

5. **SCD Type 2 Table Maintenance**

- Update the `ordersummary` table whenever there is a change in the `customer` table.
- Make old records inactive and new records active, and update `StartDate` and `EndDate` accordingly.

6. **Customer Aggregate Spend Table Creation**

- Create a `customeraggregatespend` table with columns `Name`, `TotalAmount`, and `Date` in `catalog="gen\_ai\_poc\_databrickscoe"` and `schema="sdlc\_wizard"` if it does not exist.

7. **Data Aggregation**

- Aggregate the `TotalAmount` column from the `ordersummary` table and group by the `Name` and `Date` columns.
- Load the aggregated data into the `customeraggregatespend` table.

- Functional Requirements:**

1. The system should be able to read CSV data from the specified volumes.
2. The system should be able to load data into Delta tables.
3. The system should be able to perform data cleansing (remove "Null"/Null and duplicate records).
4. The system should be able to create the `ordersummary` table if it does not exist.
5. The system should be able to join `customer` and `order` data using the `CustId` field.
6. The system should be able to load joined data into an SCD Type 2 table.
7. The system should be able to update the `ordersummary` table whenever there is a change in the `customer` table.
8. The system should be able to create the `customeraggregatespend` table if it does not exist.
9. The system should be able to aggregate data from the `ordersummary` table and load it into the `customeraggregatespend` table.

- Non-Functional Requirements:**

1. The system should ensure data consistency and integrity.
2. The system should be able to handle large volumes of data.
3. The system should be able to perform data processing in a timely manner.

- Assumptions and Dependencies:**

1. The input CSV files are in the correct format and are located in the specified volumes.
2. The Delta tables are created with the correct schema.
3. The `ordersummary` and `customeraggregatespend` tables are created with the correct schema.

- Acceptance Criteria:**

1. The system successfully ingests data from the input CSV files.
2. The system performs data cleansing correctly.
3. The `ordersummary` table is created with the correct schema.
4. The system joins `customer` and `order` data correctly.

5. The `ordersummary` table is updated correctly when there are changes in the `customer` table.

6. The `customeraggregatespend` table is created with the correct schema.

7. The system aggregates data correctly and loads it into the `customeraggregatespend` table.

- Risks and Mitigation Strategies:**

1. Data quality issues: Implement data validation and cleansing to mitigate this risk.

2. Performance issues: Optimize data processing and use efficient data storage solutions to mitigate this risk.

By following this BRD, the development team should be able to design and implement a data integration and aggregation process that meets the business requirements.